MADHU SHYAM SUNDAR REDDY KOTA

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SUMMARY

Certified Data Engineer with experience designing and maintaining scalable data pipelines using PySpark, SQL, Databricks, and AWS. Skilled in building cloud-native data platforms, optimizing data processing workflows through Delta Lake and serverless architectures, and implementing robust data validation frameworks. Experienced in Generative AI and LLM-based analytics with hands-on expertise in Amazon Bedrock, LangChain, and RAG. Proven ability to reduce infrastructure costs, automate manual processes, and deliver maintainable solutions that support data-driven decision-making across manufacturing and analytics domains.

TECHNICAL SKI LLS

Cloud & DevOps

- AWS Services: S3, EMR, Lambda, DynamoDB, Redshift, CloudWatch, AppRunner, Amplify
- GCP Services: Cloud Run, Artifact Registry, Cloud Build, IAM, BigQuery, Firestore, Cloud Storage
- DevOps & CI/CD: GitHub Actions

Data Engineering

- PySpark, Delta Lake, Databricks, SQL

Generative AI

- Amazon Bedrock (Nova), Vertex AI, Vertex AI Vector Search, LangChain, RAG

Programming & Development

- Languages: Python, SQL
- Tools & Frameworks: FastAPI, Pandas, PyYAML

EXPERIENCE

Jan 2024 - Present

Software Engineer, Info Services

INAAS (Insight-as-a-Service) – AI Analytics Agent

- Developed an AI-powered analytics agent that converts natural language questions into SQL queries and delivers real-time insights, enabling non-technical users to query data warehouses without writing code
- Achieved 90% query accuracy with response times under 30 seconds for standard datasets, leveraging Amazon Bedrock (Nova) with LangChain and RAG for context-aware SQL generation
- Designed a multi-data warehouse query engine supporting Databricks SQL and PostgreSQL through a unified abstraction layer, with S3 as a data lake backend
- Implemented session-aware memory using DynamoDB, enabling multi-turn analytical conversations where the agent remembers context across queries
- Researched and applied RAG architecture and prompt engineering techniques to train the LLM on schema understanding and complex query generation, overcoming challenges in handling diverse user question patterns
- Deployed a serverless production-ready architecture using AWS AppRunner (backend) and AWS Amplify (frontend), currently being demoed to prospective clients

Tech Stack: Python, SQL, LangChain, Amazon Bedrock (Nova), Databricks SQL, PostgreSQL, AWS S3, DynamoDB, RAG, AWS AppRunner, AWS Amplify

Data Migration Engineer – Oracle to BigQuery Modernization
(Client/Internal Project)

- Worked on migrating enterprise data workflows from Oracle to Google BigQuery, redesigning existing ETL pipelines and adapting transformation logic for the GCP ecosystem.
- Supported migration of 50-60 TB of historical and incremental data from on-prem/Oracle systems to Google Cloud Platform (GCP).
- Developed and modified workflows for loading, transforming, and validating large-scale datasets in BigQuery.
- Utilized GCP services including BigQuery, Cloud Run, Google Cloud Storage (GCS), IAM, and Cloud Logging to build scalable cloud-native data solutions.
- Implemented optimized data ingestion and storage strategies to improve query performance and reduce processing overhead in BigQuery.
- Collaborated with cross-functional teams to ensure schema compatibility, data validation, and successful migration with minimal downtime.
- Automated deployment and monitoring processes using CI/CD and cloud-native tooling.

Tech Stack: BigQuery, GCP, Cloud Run, GCS, SQL, Python, Data Migration, ETL, Oracle, IAM, Cloud Logging

Harrys Inc. – Data Engineer (Manufacturing Analytics)

- Built an end-to-end data pipeline to centralize and analyze razor manufacturing data, enabling data-driven insights for production optimization, quality control, and operational decision-making
- Engineered a medallion architecture (Raw → Silver → Gold) using Delta Lake, processing 2GB daily from 4 source types (CSV, ZIP, PDF) via PySpark on EMR Serverless
- Built 15 data assets (7 dimension tables, 8 fact tables, 5 stored procedures, 10 views) in Redshift, powering 4 Power BI dashboards for 3 business analysts
- Implemented data validation rules to filter junk records and handle schema drift, automating 10-20 hours/week of manual Excel-based analysis
- Optimized processing by splitting monolithic tables into 2 specialized fact tables, reducing query complexity and improving join performance
- Reduced infrastructure costs using Delta Lake for S3 storage optimization and EMR Serverless for on-demand processing with GitHub Actions CI/CD

Tech Stack: PySpark, AWS EMR Serverless, S3, Delta Lake, Redshift, SQL, Power BI

Synthetic Data Generator – Open Source Project

- Built a zero-cost LLM-powered CLI tool to generate realistic synthetic CSV data from YAML schemas, solving the challenge of lacking test data for pipeline development
- Generates 10K records/minute using free open-source LLM APIs (Groq, Hugging Face) with validation support (unique, not null) and reference dataset integration
- Implemented a pluggable adapter pattern for token-safe prompt generation and automated deployment via GitHub Actions
- Used in personal projects to generate mock datasets for testing data pipelines and demos, eliminating dependency on production data

Tech Stack: Python, LangChain, Groq, Hugging Face, GitHub Actions, PyPI

CERTIFICATIONS

AWS Certified Cloud Practitioner
Databricks Certified Data Engineer Associate
GCP Associate Data Practitioner Certification

AWS Certified Data Engineer Associate
Databricks Certified Generative AI Engineer Associate
GCP Generative AI Leader Certification

EDUCATION

B.Tech – Electronics and Communication Engineering

Kalasalingam Academy of Research and Education - 8.67 CGPA